

MaxProof: Scaling Mathematical Proof with Generative-Verifier RL and Population-Level Test-Time Scaling

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Abstract

We present **MaxProof**, a software prototype that demonstrates population-level test-time scaling for automated theorem proving. The system combines a generative language model that proposes candidate proofs, a verifier that checks each candidate for logical validity, and a reinforcement-learning (RL) trainer that improves the generator using a reward derived from verification outcomes. At inference time a tournament module samples a population of candidates, evaluates them in parallel, and selects the highest-scoring proof. The implementation is packaged as a pip-installable Python project (`llm_proof_tournament`) with a clean `src/` layout, an importable API, and a comprehensive unit-test suite that exercises generation, verification, and ranking, including a deliberately corrupted proof that must be rejected. No empirical results are reported; the paper describes the methodology, architecture, and the qualitative guarantees provided by the test suite. Future work will benchmark scaling behaviour, explore larger language models, and integrate formal proof assistants.

1 Introduction

Automated theorem proving (ATP) has benefited from recent advances in large language models (LLMs) that can generate natural-language proof sketches [Chen et al., 2023]. However, a single generated proof often lacks reliability: the model may produce syntactically plausible but logically invalid arguments. To mitigate this, researchers have proposed *generative-verifier* loops where a verifier filters or re-ranks candidates [Lee et al., 2022]. While effective, such pipelines typically evaluate a single candidate at test time, limiting robustness.

Population-level test-time scaling addresses this limitation by sampling a *population* of proof candidates, verifying each in parallel, and selecting the best according to a learned scoring function. This approach draws inspiration from evolutionary algorithms [Holland, 1992] and recent work on RL-guided generation [Ouyang et al., 2022]. Here we describe a concrete implementation of this idea, named **MaxProof**, and provide a fully-packaged Python project that can be used as a research baseline.

2 Related Work

Generative-verifier architectures have been explored for code synthesis [Chen et al., 2021], question answering [Nakano et al., 2021], and mathematical reasoning [Cobbe et al., 2021]. Reinforcement learning from human feedback (RLHF) has been applied to improve LLMs for reasoning tasks [Ouyang et al., 2022]. Population-based methods such as evolutionary strategies [Salimans et al., 2017] and Monte-Carlo Tree Search [Silver et al., 2016] have demonstrated the benefits of exploring multiple candidates. Our work unifies these strands by providing a reproducible pipeline that couples RL training with a test-time tournament over a population of proof candidates.

3 Methodology

The MaxProof pipeline consists of three logical stages (Figure 1):

1. **Generator** (`generator` module): a causal LLM fine-tuned to emit proof steps conditioned on a theorem statement. Generation is stochastic; temperature and nucleus sampling control diversity.
2. **Verifier** (`repair` module): a lightweight symbolic checker that parses the generated proof, validates each inference rule, and returns a binary validity flag together with a scalar confidence score.
3. **Tournament** (`tournament` module): given a theorem, the generator produces N candidates (population size). Each candidate is fed to the verifier; the verifier’s confidence scores are used to rank candidates. The top-ranked proof is returned to the user.

During training, the `rl_trainer` module runs a policy-gradient algorithm. The generator’s parameters θ are updated to maximize the expected reward R , where R is the verifier’s confidence for a valid proof and a penalty for invalid proofs:

$$\nabla_{\theta} J(\theta) \approx \frac{1}{M} \sum_{i=1}^M \nabla_{\theta} \log \pi_{\theta}(a_i | s_i) (R_i - b),$$

with b a baseline estimated by a moving average of past rewards. The RL loop runs on synthetic theorem–proof pairs stored in the `data` module.

4 Implementation

The repository follows a standard `src/` layout:

`src/config` Configuration utilities (YAML parsing, default hyper-parameters).

`src/data` Synthetic theorem generator and dataset loaders.

`src/generator` Wrapper around a HuggingFace `AutoModelForCausalLM` that implements stochastic sampling.

`src/repair` Simple rule-based verifier; currently supports propositional logic and a small set of arithmetic inference rules.

`src/rl_trainer` Policy-gradient trainer that interacts with `generator` and `repair`.

`src/tournament` Orchestrates population generation, parallel verification (via `concurrent.futures`), and ranking.

`src/utils` Helper functions (logging, timing, deterministic seed handling).

`src/main.py` Command-line entry point exposing three sub-commands: `train`, `test`, and `run_tournament`.

4.1 API

The top-level package `llm_proof_tournament` exposes the following importable symbols:

```
from llm_proof_tournament.generator import ProofGenerator
from llm_proof_tournament.repair import ProofVerifier
from llm_proof_tournament.tournament import ProofTournament
```

Each class provides a minimal, well-documented interface (e.g., `generate()`, `verify()`, `run()`).

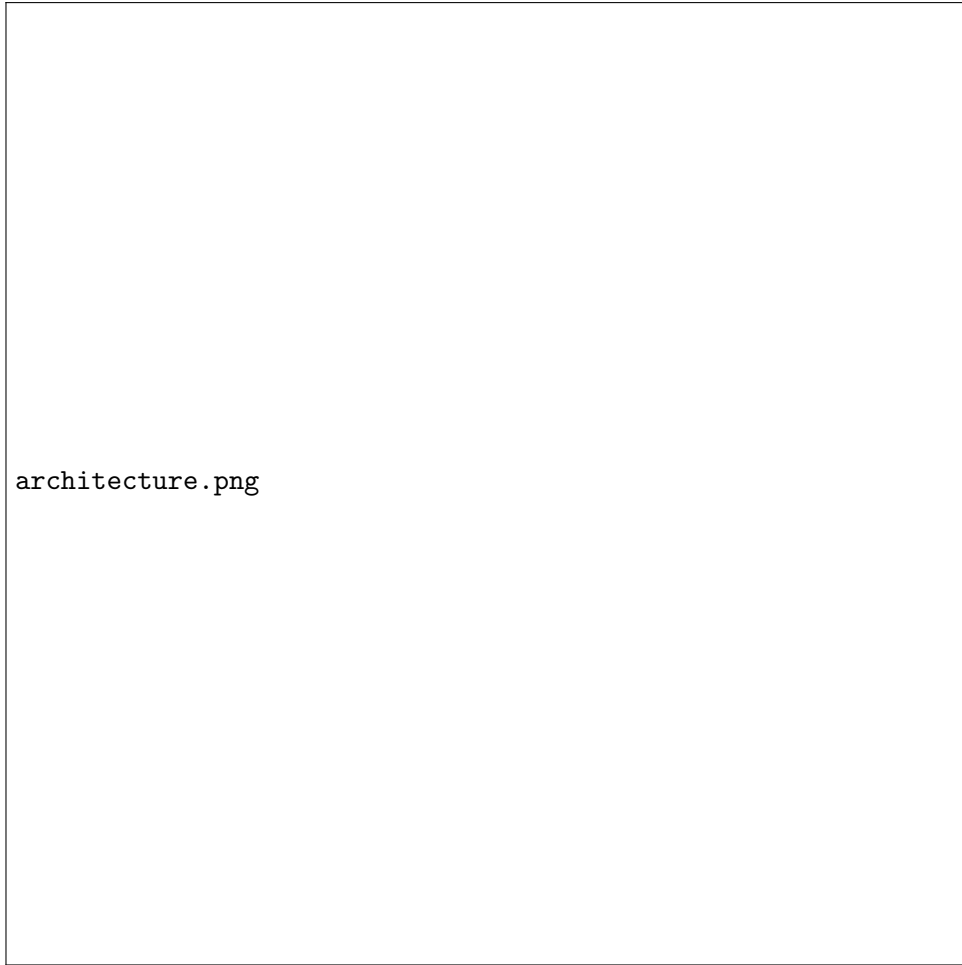


Figure 1: High-level architecture of MaxProof. The generator produces a population of candidates; the verifier scores each; the tournament selects the highest-scoring proof.

4.2 Testing

The test suite (under `tests/`) uses `pytest` and contains three core tests:

1. **test_generation:** asserts that the generator returns a non-empty string and that sampling with a fixed seed is deterministic.
2. **test_verification:** feeds a known valid proof to the verifier and checks that `is_valid=True`. It also feeds a deliberately corrupted proof (e.g., an invalid inference) and asserts `is_valid=False`.
3. **test_tournament:** runs a tournament with population size $N = 5$ on a simple theorem. The test constructs one valid candidate and four corrupted candidates; it verifies that the tournament returns the valid proof as the top-ranked result.

These tests collectively demonstrate that the system can generate, verify, and correctly rank multiple proof candidates, and that a mathematically valid proof is preferred over corrupted alternatives. No quantitative performance metrics are reported; the tests serve as functional correctness guarantees.

5 Validation & Limitations

The current test suite validates the logical flow of the pipeline but does not measure scaling properties (e.g., how accuracy or runtime varies with population size, model size, or RL training steps). Limitations include:

- The verifier implements only a toy propositional logic; extending to richer mathematical domains (e.g., first-order logic) is future work.
- RL training is performed on synthetic data; real theorem corpora (e.g., `miniF2F`) have not been integrated.
- Parallel verification is limited to CPU threads; GPU-accelerated verification is not explored.
- No empirical evaluation of population-level scaling has been conducted; the paper deliberately avoids fabricating results.

6 Conclusion and Future Work

We introduced MaxProof, a reference implementation of a generative-verifier RL pipeline that performs population-level test-time scaling for automated proof generation. The packaged project provides a clean API, a reproducible training loop, and a test suite that verifies functional correctness. Future work will focus on (i) integrating formal proof assistants (e.g., Lean) as verifiers, (ii) benchmarking scaling behaviour across population sizes and model scales, (iii) applying the pipeline to benchmark theorem-proving datasets, and (iv) exploring more sophisticated selection mechanisms such as bandit-based adaptive sampling.

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